Aether

Data Analysis Report

Empirical diagnostic evaluation, descriptive statistics, and multivariate exploratory analysis.

DATASET NAME	**Quarterly_Sales_Performance.csv**
REPORT DATE	07 October 2026, 20:23 UTC
DATASET VERSION	Version 1
TOTAL RECORDS	150 rows
FEATURE DIMENSIONS	8 columns (4 numeric, 2 categorical)
DATA QUALITY SCORE	**99/100**
SCOPE OF ANALYSIS	Full-Spectrum Exploratory & Statistical Audit

Notice: This report is generated dynamically from verified local dataset records. All numerical metrics, distribution profiles, and correlation coefficients are computed strictly from the active dataset version.

Executive Summary

TOTAL RECORDS	TOTAL DIMENSIONS	DATA QUALITY SCORE
150	**8**	**99/100**
MISSING DATA RATE	DUPLICATE RECORDS	DETECTED OUTLIERS
0.33%	**2**	**20**

This report presents a structured analysis for **Quarterly_Sales_Performance.csv** (Version 1). The dataset comprises **150 observations** across **8 attributes**. Empirical evaluation yields an overall Data Quality Score of **99/100**, measuring record completeness (100%), uniqueness (97%), and validity (100%).

Key Summary Findings:

• The dataset contains 150 verified observations across 8 attributes with an overall Data Quality Score of 99/100.

• Data completeness is 100%, with 4 missing values concentrated in ['Customer_Rating'].

• Strongest linear association exists between 'Sales_Amount' and 'Customer_Rating' (r = -0.06, little or no linear relationship).

• Categorical concentration observed in 'Region' where 'North America' accounts for 45.33% of total entries.

Dataset Overview

DATASET NAME	**Quarterly_Sales_Performance.csv**	ANALYSIS DATE	07 October 2026, 20:23 UTC
ROW COUNT	150	COLUMN COUNT	8
FILE FORMAT	CSV	MEMORY FOOTPRINT	31.5 KB

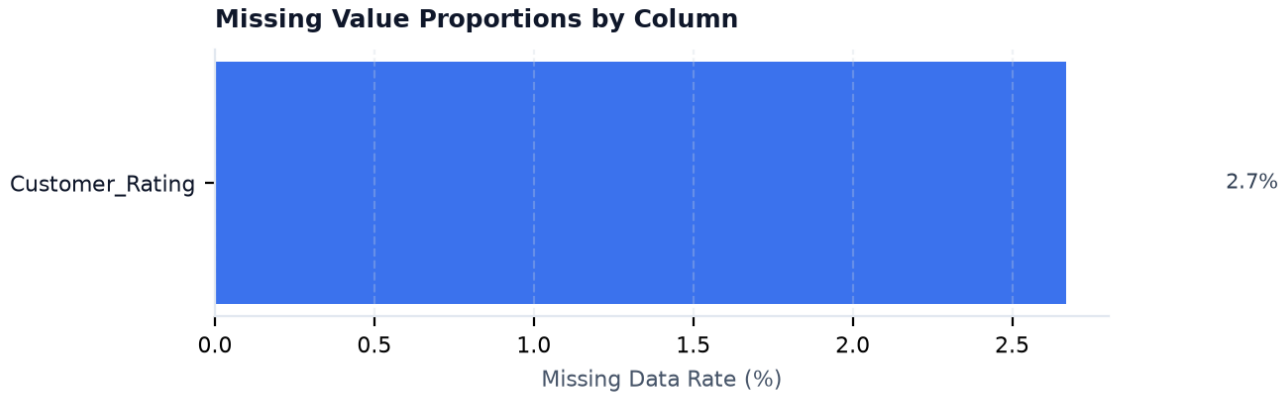
Column Overview:

Column	Data Type	Non-null	Missing	Unique	Summary
Transaction_ID	str	150	0 (0.0%)	148	148 unique identifiers
Date	datetime64[us]	150	0 (0.0%)	148	Temporal sequence
Region	str	150	0 (0.0%)	4	Top: North Americ (45%)
Category	str	150	0 (0.0%)	4	Top: Office Suppl (35%)
Sales_Amount	float64	150	0 (0.0%)	148	[57.2 to 4500.0], Med: 417.3
Profit_Margin	float64	150	0 (0.0%)	148	[0.0 to 0.4], Med: 0.2
Units_Sold	int32	150	0 (0.0%)	19	[1.0 to 19.0], Med: 10.0
Customer_Rating	float64	146	4 (2.67%)	5	[1.0 to 5.0], Med: 4.0

Data Quality

Quality Dimension	Score	Benchmark	Assessment Detail
Completeness	**100%**	>= 95.0%	Calculates proportion of non-null cells across the observation matrix.
Uniqueness	**97%**	>= 98.0%	Measures absence of identical duplicate row vectors.
Validity	**100%**	>= 90.0%	Evaluates presence of constant zero-variance features.
Consistency	**100%**	>= 85.0%	Checks for type homogeneity and uniform data formatting.

Audit Details: 4 missing cells identified (0.33% of dataset). Duplicate row analysis identified 2 duplicate record(s) (1.33%).

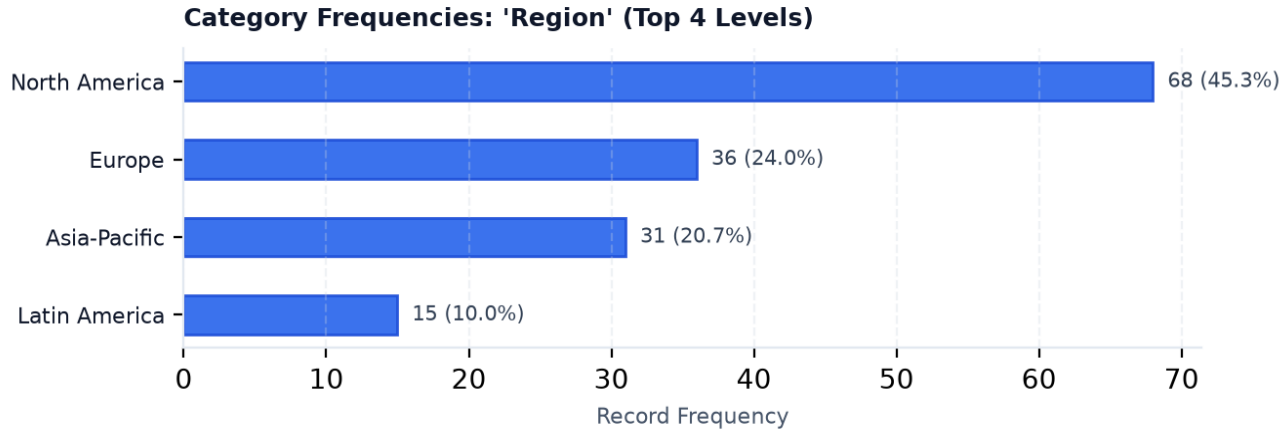


Descriptive Statistics

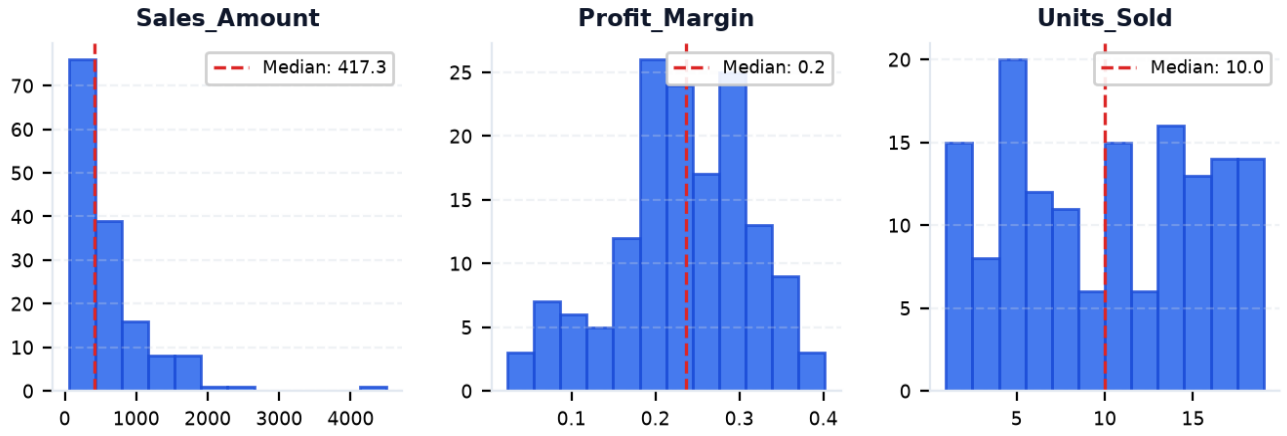
Feature	Count	Mean	Median	Std Dev	Min	Max	Q1	Q3	IQR	Skew
Sales_Amount	150	581.0	417.3	579.6	57.2	4500.0	197.5	723.6	526.2	2.81
Profit_Margin	150	0.2	0.2	0.1	0.0	0.4	0.2	0.3	0.1	-0.36
Units_Sold	150	9.9	10.0	5.5	1.0	19.0	5.0	15.0	10.0	0.04
Customer_Ratin	146	4.0	4.0	0.9	1.0	5.0	4.0	5.0	1.0	-1.14

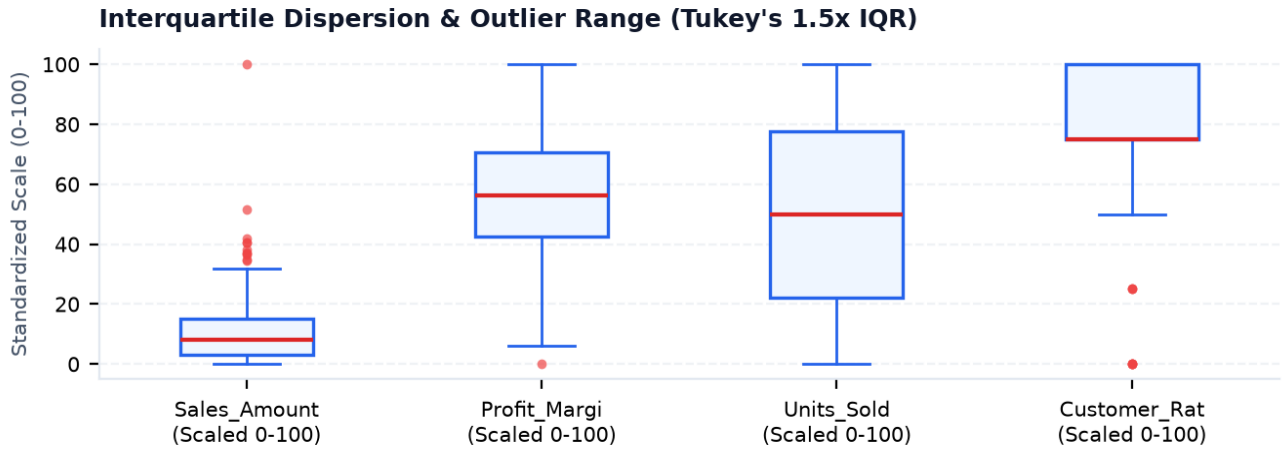
Categorical Analysis

Column	Unique	Most Frequent Value	Frequency	Percentage	Status
Region	4	North America	68	45.33%	Dominant (45.33%)
Category	4	Office Supplies	53	35.33%	Balanced

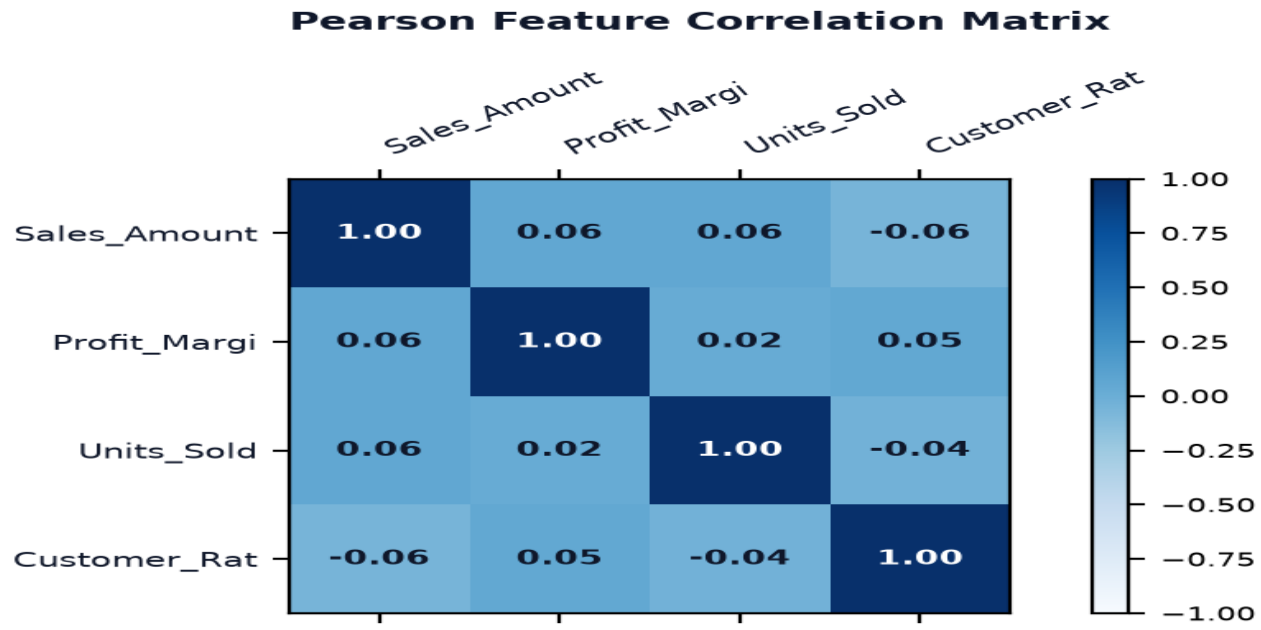


Distribution Analysis



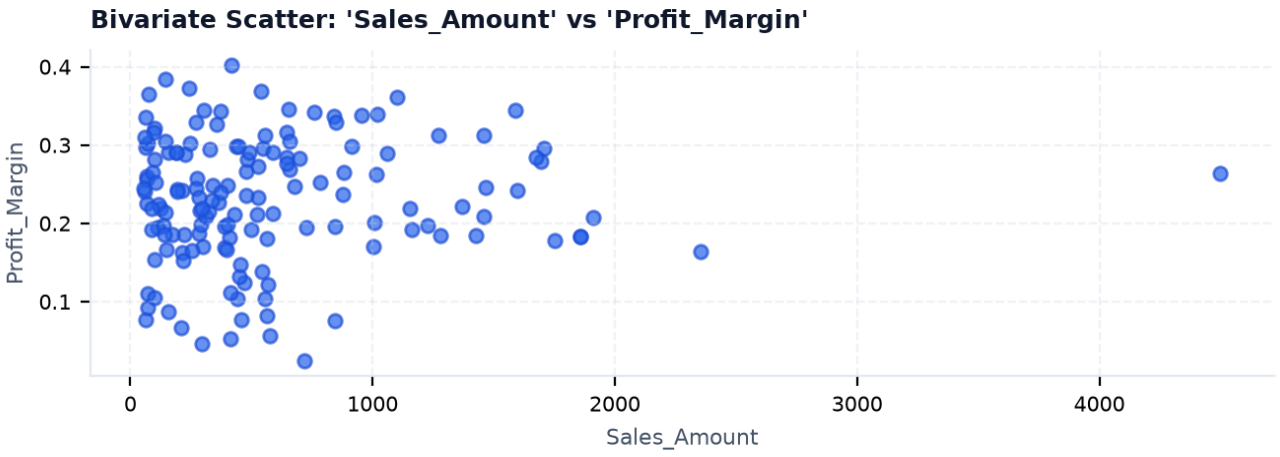


Correlation Analysis



Feature X	Feature Y	Pearson r	Strength	Direction
Sales_Amount	**Customer_Rating**	-0.06	Little or no linear relationship	Positive
Sales_Amount	**Profit_Margin**	0.06	Little or no linear relationship	Positive
Sales_Amount	**Units_Sold**	0.06	Little or no linear relationship	Positive
Profit_Margin	**Customer_Rating**	0.05	Little or no linear relationship	Positive
Units_Sold	**Customer_Rating**	-0.04	Little or no linear relationship	Positive

Relationships



Outlier Analysis

Feature	Outlier Count	Percentage	Lower Bound (1.5x IQR)	Upper Bound (1.5x IQR)
Sales_Amount	11	7.33%	-591.75	1512.87
Profit_Margin	1	0.67%	0.03	0.45
Units_Sold	0	0.0%	-10.00	30.00
Customer_Rating	8	5.48%	2.50	6.50

Key Findings

• The dataset contains 150 verified observations across 8 attributes with an overall Data Quality Score of 99/100.

• Data completeness is 100%, with 4 missing values concentrated in ['Customer_Rating'].

• Strongest linear association exists between 'Sales_Amount' and 'Customer_Rating' (r = -0.06, little or no linear relationship).

• Categorical concentration observed in 'Region' where 'North America' accounts for 45.33% of total entries.

• Temporal analysis indicates an overall stable volume with peak activity recorded during 2026-01.

Recommendations

Category	Priority	Action Recommendation
Analytical	**Medium**	Apply robust scaling or segment upper-quartile outliers in 'Sales_Amount' to prevent extreme-value distortion.
Operational	**Low**	Monitor the dominant category 'North America' in 'Region' to identify potential sampling bias or operational dependency. Further domain validation is recommended.
Operational	**Medium**	Align operational capacity and staffing with peak periods observed in 2026-01. Further domain validation is recommended.
Data Quality	**Low**	Maintain automated schema validation checks on future dataset ingestion to ensure consistent type mappings.

Appendix

Mathematical Methodology	• **Tukey's IQR Outlier Rule:** Outliers identified strictly outside [Q1 - 1.5*IQR, Q3 + 1.5*IQR]. • **Pearson Correlation:** Linear association computed as r = cov(X, Y) / (std(X) * std(Y)). • **Fisher-Pearson Skewness:** g1 = m3 / (m2^(3/2)) measuring distribution symmetry.
Data Quality Scoring	Overall score computed via weighted evaluation: Completeness (35%) + Uniqueness (25%) + Validity (20%) + Consistency (20%). Penalties scale directly with missing cell frequency and duplicate rows.
Compilation Provenance	Generated by Aether Engine • ReportLab PDF Architecture • Python 3.0.5 / NumPy 2.5.2. Timestamp: 07 October 2026, 20:23 UTC.